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Receipt OCR Benchmark

Comparing traditional OCR with Azure AI solutions

Results from 6 mock medical receipts · October 2026

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Objective and Methodology

We tested four approaches on the same receipts, fields and scoring to support a build-or-buy decision.

BUSINESS QUESTION

Is Azure Document Intelligence + Azure OpenAI worth its additional cost and complexity?

Compared with simpler OCR solutions that run on our own compute.

Dataset

- 6 mock medical receipts in 4 layouts
- 4 handwritten, 2 printed

Fields evaluated

- Doctor name, patient name, receipt number
- Service date, total amount, registration number

Approaches compared

- PaddleOCR Light and Heavy
- Azure DI, and Azure DI + Azure OpenAI

Evaluation criteria

- Accuracy · Speed · Reliability · Cost

Every receipt × field was marked correct or wrong against one hand-checked answer key (36 cells; 25 have a value on the receipt).

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Four Approaches Compared

Each approach reads the receipt (OCR), then finds the six fields; they differ in where each step runs.

	STEP 1 · OCR: read the text	STEP 2 · Extract the six fields	RESULT
1 **PaddleOCR Light**	**PaddleOCR Light** Open-source OCR, small model	**General rules** Same fixed rules	**6 fields**
2 **PaddleOCR Heavy**	**PaddleOCR Heavy** Open-source OCR, large model	**General rules** Same fixed rules	**6 fields**
3 **Azure DI**	**Azure Document Intelligence** Cloud OCR (prebuilt-read)	**General rules** Same fixed rules	**6 fields**
4 **Azure DI + AOAI**	**Azure Document Intelligence** Cloud OCR (prebuilt-read)	**Azure OpenAI (gpt-5-mini)** LLM reads the text	**6 fields**

Runs on our own compute (no per-call fee)

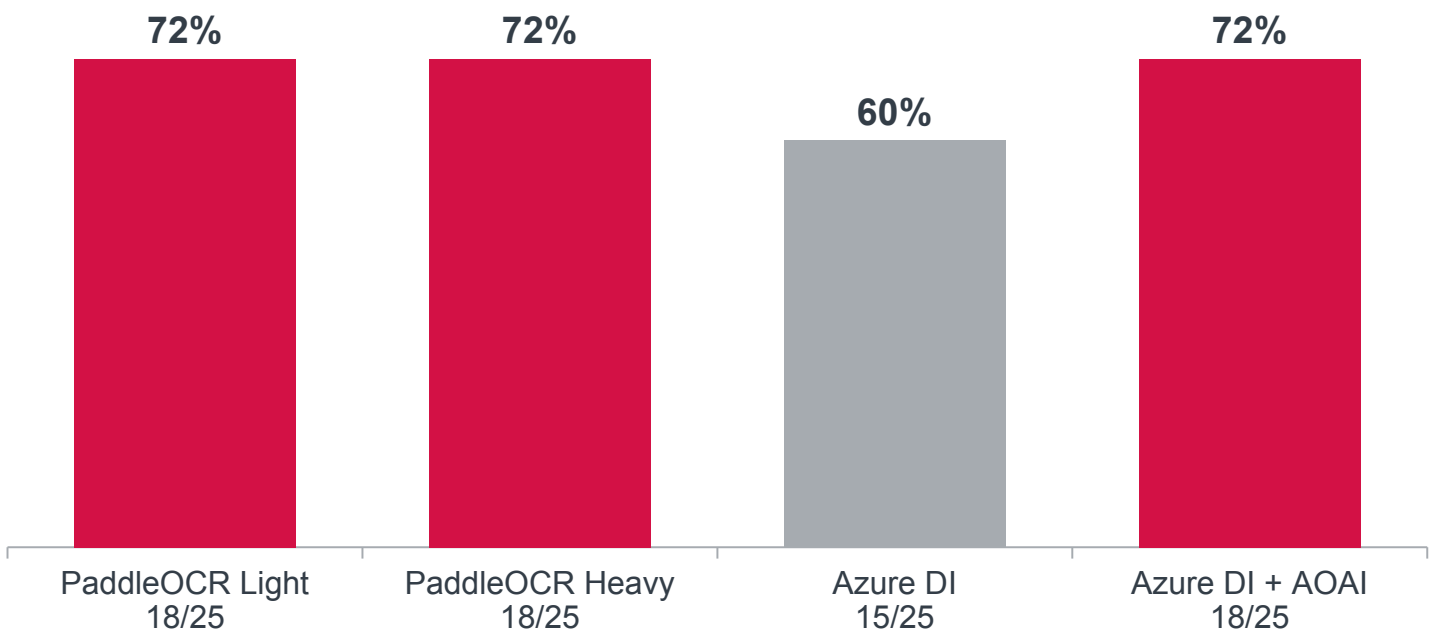
Paid Azure API call

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Accuracy Comparison

Three approaches tie at 72% accuracy; Azure DI with rules alone is lowest at 60%.



Based on 6 mock receipts: one field = 4 percentage points, so small differences are not conclusive.

HOW ACCURACY IS MEASURED

Correct fields ÷ fields that exist on the receipts

- 25 of 36 receipt-field cells have a value; fields not on a receipt are excluded
- A missing value or a failed receipt counts as wrong
- Values given for fields not on the receipt are counted separately: 1 for Azure DI + AOAI, 0 for the others

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Accuracy by Field

Totals are solved by all four; Azure OpenAI finds more names and numbers, PaddleOCR reads handwritten dates.

Field	PaddleOCR Light	PaddleOCR Heavy	Azure DI	Azure DI + AOAI
Total amount	6/6	6/6	6/6	6/6
Patient name	2/4	2/4	2/4	**4/4**
Receipt number	2/3	2/3	2/3	**3/3**
Doctor name	3/6	3/6	**4/6**	**4/6**
Service date	**5/6**	**5/6**	1/6	1/6
Registration number	*Not on any receipt, so not scored*			

Highlighted = best result for that field. Cells show correct / receipts where the field exists.

WHAT STANDS OUT

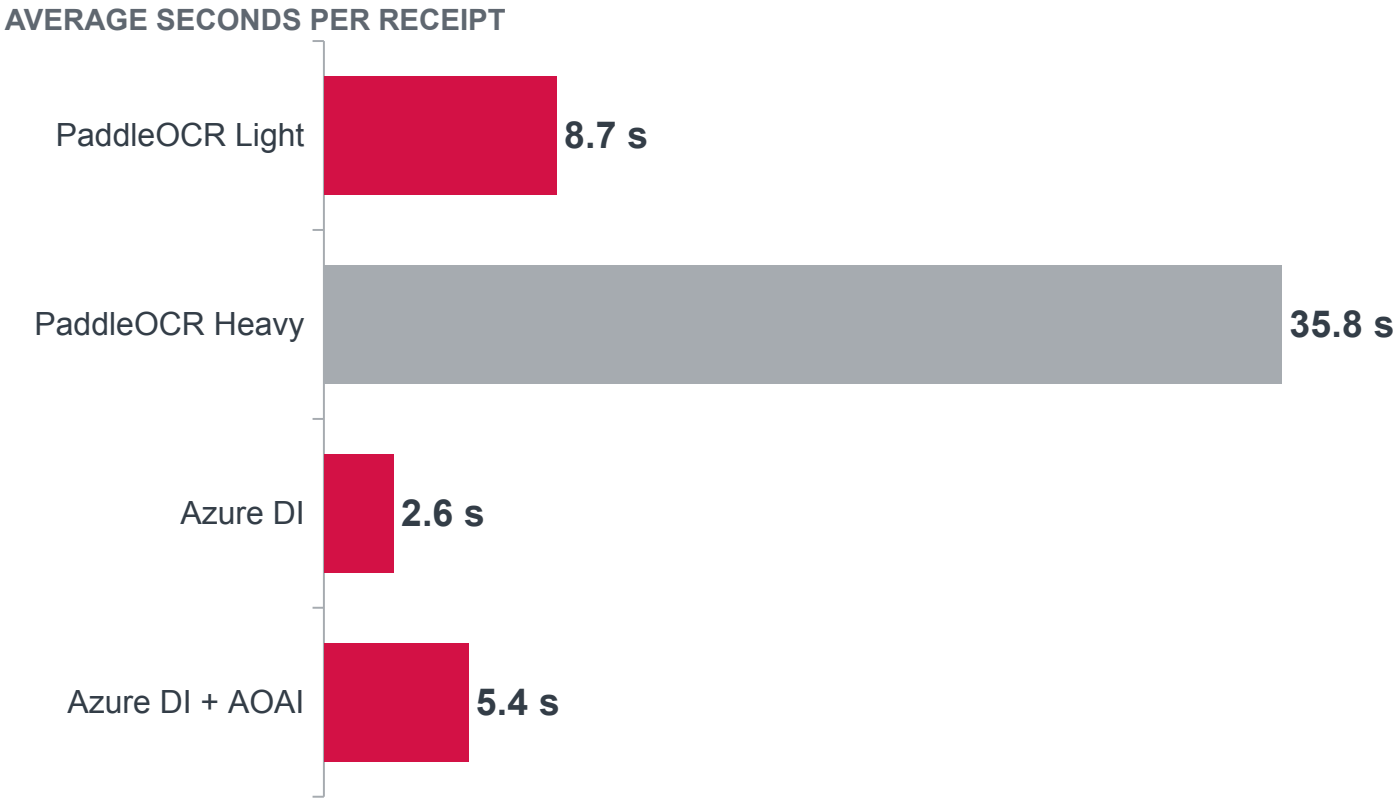
- Total amount: correct on all receipts for all four approaches
- Azure DI + AOAI found all 4 patient names and all 3 receipt numbers
- PaddleOCR read handwritten dates correctly (5/6)
- Both Azure DI approaches: 1/6 dates, because Azure DI read the handwritten year 2026 as 2020

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Processing Speed and Reliability

Azure DI is fastest; every approach processed all 6 receipts without errors.



RELIABILITY

6 / 6

receipts processed by every approach, no errors or timeouts

- PaddleOCR Heavy needed a library version fix to run on Azure CPU
- Older Azure OpenAI setup timed out; the final version took 3 s per call

Times reflect different methods: PaddleOCR = CPU time on Azure ML compute; Azure = API call incl. network. One receipt at a time; not measured at volume.

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Main Findings and Cost Considerations

Azure OpenAI lifts Azure DI from 60% to 72%, but PaddleOCR Light reaches the same accuracy without API calls.

1. **Heavy adds nothing**
   PaddleOCR Heavy matched Light at 72% but was about 4× slower.

2. **Azure OpenAI helps DI**
   Accuracy rose from 60% to 72% over Azure DI alone.

3. **Better on unclear layouts**
   The LLM found names and numbers the fixed rules missed.

4. **Limited by DI's OCR**
   4 of its 7 errors came from DI misreading handwritten years.

5. **Light matches it**
   PaddleOCR Light: 72% with no paid API calls (compute still costs).

COST DRIVERS PER 1,000 RECEIPTS (MEASURED)

Approach	Usage	Price
PaddleOCR Light	2.4 VM-hours	**Pending**
PaddleOCR Heavy	9.9 VM-hours	**Pending**
Azure DI	1,000 DI pages	**Pending**
Azure DI + AOAI	1,000 DI pages + 0.55M input / 0.28M output tokens	**Pending**

Prices needed: Azure ML VM hourly rate, Document Intelligence Read price per 1,000 pages, gpt-5-mini price per 1M tokens.

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Conclusion and Recommendation

Keep two candidates and test them on a larger, representative set before choosing a production solution.

KEEP AS CANDIDATES

PaddleOCR Light **72% · 8.7 s**

No API fees, runs offline, best on handwritten dates

Azure DI + AOAI **72% · 5.4 s**

Best on names and receipt numbers; paid per receipt

WEAKER ON THIS DATASET

PaddleOCR Heavy **72% · 35.8 s**

Same accuracy as Light, about 4× slower

Azure DI alone **60% · 2.6 s**

Fastest, but the lowest accuracy

Important: 6 mock receipts are not enough to select a production solution.

NEXT STEP 1

Test on more receipts

Run both candidates on a larger, representative set of real, anonymised receipts

NEXT STEP 2

Confirm pricing

VM rate, Document Intelligence page price and gpt-5-mini token prices

NEXT STEP 3

Optional experiment

PaddleOCR Light + Azure OpenAI: an idea, not yet tested

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